# PKM2-deficiency induced fatty acid biosynthesis activation limits the response of non-small cell lung cancer to Osimertinib

Experimental dataset evidence card

Document ID: DATASET-GSE300311
Category: experimental_datasets
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## Dataset Identity

Dataset ID: GSE300311
Title: PKM2-deficiency induced fatty acid biosynthesis activation limits the response of non-small cell lung cancer to Osimertinib
Taxon: Homo sapiens
Sample count: 16
Selected reason: PC9 PKM2-deficient/control samples with vehicle/osimertinib and chromatin marks

## Experimental Context

Assay context: resistance model; EGFR inhibition; osimertinib response
Sample accessions: GSM9058183; GSM9058175; GSM9058172; GSM9058178; GSM9058184; GSM9058181; GSM9058170; GSM9058176; GSM9058173; GSM9058179; GSM9058174; GSM9058171; GSM9058185; GSM9058177; GSM9058182; GSM9058180
Summary: Non-small cell lung cancer (NSCLC) - the most common form of lung cancer - is an aggressive malignancy with limited effective therapeutic options1. Osimertinib, a third-generation epidermal growth factor receptor tyrosine kinase inhibitor (EGFR-TKI), has revolutionized treatment for EGFR-mutant NSCLC, yet resistance significantly limits its clinical benefit to patients2. Here, using a genome-wide genetic screen, we discover that pyruvate kinase M2 (PKM2) is a novel regulator of Osimertinib resistance. Loss of PKM2 triggers fatty acid biosynthesis activation in a glycolysis-independent manner. We have shown that PKM2 interacts with KDM1A in cytosol and nuclear translocation of KDM1A is increased in both PKM2-KO and Osimertinib-persister cells. Mechanistically, KDM1A demethylates H3K9me3 in the promoter region of the key lipid synthesis enzyme, fatty acid synthase (FASN), and cooperates with transcription factor AP-2 to activate FASN transcription, thereby conferring Osimertinib resistance. Pharmacological inhibition of FASN by Orlistat induces synthetic lethality with Osimertinib in vitro and vivo Osimertinib-resistant models. Treatment of 9 NSCLC patients who were unresponsive t...

## Provenance

Provenance: derived_from_raw_layer
Privacy posture: cell-line or assay-level source selected
Raw source trace: 00_raw/_corpus/json/datasets/geo/GSE300311/geo_esummary.json; 00_raw/_corpus/txt/datasets/geo/GSE300311/series_soft.txt; 00_raw/_corpus/json/datasets/geo/GSE300311/source_record.json